

Industrial Safety Intelligence Platform

AI-Powered Compound Risk Detection for Zero-Harm Operations

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Abstract

Industrial accidents continue to cause significant human and financial losses due to delayed risk identification and fragmented safety monitoring systems. Existing industrial environments generate large amounts of data through IoT sensors, work permits, worker tracking systems, SCADA platforms, and regulatory compliance records. However, these data streams are typically analyzed independently, resulting in missed compound risks and delayed interventions.

This project presents an AI-Powered Industrial Safety Intelligence Platform that integrates multiple industrial data sources into a unified safety intelligence framework. The platform utilizes compound risk analysis, trend detection, lead-time prediction, and risk scoring to identify hazardous situations before they escalate into critical incidents. The proposed solution transforms industrial safety management from reactive incident response to proactive risk prevention.

1 Introduction

Industrial facilities operate in highly dynamic environments where multiple factors influence workplace safety. Gas leakage, equipment malfunction, maintenance activities, worker presence, and permit violations can individually contribute to hazardous conditions. In many cases, accidents occur not because a single parameter exceeds its threshold but because several seemingly moderate risks combine to create a critical situation.

Traditional monitoring systems often focus on isolated sensor readings and predefined alarm thresholds. Such approaches fail to capture the relationships between multiple operational variables and therefore miss many dangerous scenarios.

The objective of this project is to build an intelligent safety platform capable of detecting compound risks, predicting incident escalation, and recommending proactive interventions.

2 Problem Statement

Current industrial safety systems suffer from the following limitations:

- Safety data exists in disconnected silos.

- IoT sensors, permit systems, and worker tracking platforms operate independently.
- Manual risk assessment delays critical decisions.
- Single-sensor alarms generate false positives and false negatives.
- Dangerous combinations of events often remain undetected.

As a result, safety teams receive fragmented information and lack real-time situational awareness necessary for preventing accidents.

3 Proposed Solution

The proposed Industrial Safety Intelligence Platform combines data from multiple industrial systems and performs intelligent risk assessment using a Compound Risk Detection Engine.

The platform performs:

1. Multi-source data ingestion.
2. Compound hazard identification.
3. Trend and anomaly analysis.
4. Lead-time prediction.
5. Risk scoring.
6. Intervention recommendation.
7. Regulatory compliance support.

The objective is to provide early warning signals and actionable insights before accidents occur.

4 System Architecture

The system architecture consists of three primary layers.

4.1 Layer 1: Multi-Source Data Ingestion

The platform collects information from:

- IoT Sensors
 - Gas concentration
 - Temperature
 - Pressure
- Work Permit Systems
- Worker Tracking Systems
- SCADA and Historical Records
- Regulatory Databases

4.2 Layer 2: Compound Risk Detection Engine

The intelligence layer consists of:

- Weighted Compound Rule Engine
- Trend Analysis Module
- Lead-Time Prediction Module
- Risk Scoring Engine

4.3 Layer 3: Action and Response Layer

This layer generates:

- Critical alerts
- Evacuation recommendations
- Permit suspension notices
- Entry lockdown advisories
- Compliance reports

Insert Architecture Diagram Here

5 Methodology

5.1 Compound Risk Detection

Instead of evaluating safety parameters individually, the platform analyzes relationships among multiple operational indicators.

Example rules include:

- Gas concentration + Worker Presence
- Active Permit + Hazardous Environment
- Confined Space + Elevated Gas Levels
- Maintenance Activity + High Temperature

Each rule contributes to the final risk score.

5.2 Trend Analysis

Historical sensor data is analyzed to identify:

- Rising trends
- Sudden spikes
- Rate-of-change anomalies
- Escalation patterns

Trend analysis enables early detection before thresholds are crossed.

5.3 Lead-Time Prediction

The system estimates the remaining time before a situation becomes critical.

Benefits include:

- Early warning generation
- Better resource allocation
- Faster emergency response

5.4 Risk Scoring

Risk scores are calculated using weighted aggregation.

$$[\text{RiskScore} = \sum_{i=1}^n \text{Weight}_i \times \text{Condition}_i]$$

Scores are normalized on a scale from 0 to 100.

Risk categories:

- 0–30 : Low Risk
- 31–60 : Moderate Risk
- 61–80 : High Risk
- 81–100 : Critical Risk

6 Prototype Features

The developed prototype includes:

- Real-Time Dashboard
- Geospatial Risk Heatmap
- Critical Alert Monitoring
- Lead-Time Prediction
- Compound Risk Detection
- False Negative Analysis
- Intervention Recommendation Engine
- Compliance Intelligence

7 Demonstration Scenario

A simulated industrial incident was created to validate the system.

7.1 Scenario

Zone B Storage Area:

- Gas Concentration = 88 PPM
- Workers Present = 3
- Hot Work Permit Active
- Confined Space Operation

Individually, these parameters may not trigger a critical alarm. However, when analyzed collectively, the platform identifies a severe compound risk.

The system generates:

- Critical Risk Alert
- Immediate Evacuation Recommendation
- Permit Suspension Advisory
- Entry Lockdown Notification

8 Expected Impact

The proposed platform offers significant benefits:

- Reduction in industrial accidents
- Improved worker safety
- Earlier hazard detection
- Reduced false-negative incidents
- Faster emergency response
- Enhanced regulatory compliance

9 Future Scope

Potential future enhancements include:

- Real-time IoT deployment
- Computer vision integration using CCTV feeds
- Multi-agent AI orchestration
- Predictive maintenance analytics
- Digital twin integration
- Large-scale industrial deployment

10 Conclusion

This project demonstrates the feasibility of an AI-powered industrial safety intelligence platform capable of detecting compound risks before accidents occur. By integrating multiple operational data sources and applying intelligent risk analysis, the system provides proactive safety interventions, improved situational awareness, and enhanced compliance support.

The proposed solution represents a significant step toward achieving zero-harm industrial operations through predictive safety intelligence.